

Spatiotemporal gait characteristics across the adult lifespan: Reference values from a healthy population – Analysis of the COMplete cohort study

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ABSTRACT

Background: Gait changes with aging have been investigated, but few studies have examined a wide range of gait parameters across the adult lifespan. This study aimed to investigate gait differences across age groups stratified by sex.

Methods: This cross-sectional study included 629 healthy, normal-weight (i.e., BMI < 30 kg/m²) participants from Switzerland of the COMplete cohort study, aged 20 to over 90 years. Gait metrics were assessed using an inertial measurement unit (IMU)-based gait analysis system, including speed, cycle duration variability, asymmetry, stride length, cycle duration, cadence, double support, stance (time foot is on the ground during a gait cycle), swing (time foot is in the air during a gait cycle), loading (early part of the stance phase), foot-flat (mid-stance phase when foot is flat), and pushing (late stance phase leading to toe-off) phases. Percentile curves were calculated using generalized additive models for location, scale, and shape.

Results: Gait data from 545 participants (273 men and 272 women) were analyzed. Participants were equally distributed across the seven age decades, with an average of 40 men and 40 women representing every decade. Both men and women showed a reduction in gait speed and stride length, and an increase in cycle duration variability and asymmetry with aging. Gait speed and stride length declined across the age groups, with a significant difference found in participants aged 80 to 91 compared to younger age groups.

Significance: Age-related changes in gait parameters were seen in both men and women. These may be attributed to the typical decline in muscle strength, balance, coordination, and neuromuscular function. The results of this study contribute to the understanding of gait changes throughout the lifespan and can be used for comparison with other populations and as reference values for individual patients.

1. Introduction

Walking is a complex task learned during childhood and performed almost automatically throughout life. Degenerative changes to the muscles and neuromuscular system related to aging can alter gait patterns in older adults, leading to decreased gait velocity, stride length,

joint movement, and an increased gait stiffness [1]. These changes are associated with an increased risk of falls and fear of falling [1,2], which further affect gait quality [3].

Subtle changes in gait patterns may occur over the lifespan, detectable only through sensitive assessment tools like pressure-sensitive walkways, camera-based or inertial measurement unit (IMU)-based

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gait analysis systems. Reference values for gait parameters across the lifespan help to evaluate an individual's walking ability, to assess the impact of gait on health outcomes, to screen for intrinsic risk factors for falls or disease indicators, and to evaluate clinical intervention effects.

A 2018 systematic review revealed a lack of data on spatiotemporal gait parameters across the entire lifespan, in a healthy population with most studies focusing on either older or younger adults [4]. The authors concluded that data concerning middle-aged adults are lacking (age of 30 up to the age of 59) to understand the effects of aging on gait. The only existing study covering a wide age range used the GAITrite walkway system in a healthy Asian population [5]. It has been shown that gait parameters differ between ethnic/racial groups [6,7]. To the best of our knowledge, there are no data available from Europe covering the entire adult lifespan. Comprehensive data are needed for a better understanding of when and how gait changes occur.

Sensor-based systems like the GaitUp Physilog®5 have become widely used in gait analysis due to their flexibility and capability to assess longer distances than traditional systems like GAITrite®. A recent study showed substantial agreement between the two systems regarding key gait parameters [8]. For our study, we chose the sensor-based Physilog®5 system for its feasibility, alignment with current gait analysis trends, and the ability to assess longer walking distances compared to a carpet-based system.

There is a relatively large body of literature focusing on gait in people with specific conditions [9–11]. This is likely because understanding gait changes in people with conditions like Parkinson's disease, stroke, multiple sclerosis, or osteoarthritis can provide valuable insights into the disease progression, and help identify potential interventions. However, studying gait in healthy individuals is also important, as it helps to establish normative data and reference values for various gait parameters. This can be useful for comparing gait patterns in individuals with and without specific conditions and for identifying early warning signs of potential health issues. Therefore, our study aimed to close this gap by providing an overview of spatiotemporal gait parameters in healthy Swiss adult individuals using a wearable sensor system placed on the shoes. The specific aims were: a) to investigate gait differences across age groups stratified by sex; and b) to illustrate age-related changes by percentile curves.

2. Methods

2.1. Study population and recruitment

We recruited healthy participants from the Basel area, Switzerland, within the COMpLETE cohort study comprising approximately 40 males and 40 females per age category (20–29, 30–39, 40–49, 50–59, 60–69, 70–79, and 80 + years). General invitation letters with information on the study aims and procedures were mailed to randomly selected districts in the Basel area. The sample size justification, recruitment scheme and measurements are detailed in the published study protocol [12]. Interested individuals underwent a telephone screening before scheduling tests. Participants' final eligibility was confirmed onsite on the day of examination. Prior to participation, participants provided written informed consent. Eligible individuals were healthy men and women aged 20 to 100 years with a body mass index of less than 30 kg/m². Exclusion criteria were any manifest exercise-limiting chronic diseases, such as, among others, stroke, lower-extremity artery disease, orthopedic issues, or any form of dementia.

The study was conducted at the Department of Sport, Exercise, and Health at the University of Basel, Switzerland. The Ethics Committee of Northwestern and Central Switzerland (EKNZ 2017–01451) approved the study.

2.2. Procedures

All measurements were conducted by trained assessors. We

instructed the participants to walk at their usual walking speed and specifically pointed out that they do not need to walk at maximum speed. The instruction was as follows: “Please walk at your normal pace for 20 m to the pylon, go around the pylon, and then walk back the 20 m past the line.”

The tests took place indoors in a corridor with a smooth and even surface. Height and body mass were measured to the nearest 0.5 cm and 0.1 kg, respectively.

2.3. Gait Parameters

Gait parameters were measured using an IMU-based gait analysis system (Physilog®, Mindmaze, Lausanne, Switzerland), which allows motion analysis based on two lightweight inertial sensors, attached to the dorsum of both feet, at the height of the metatarsal bones. The system has high validity and reliability [8,13,14]. An overview of measured parameters and their definitions is given in Table 1.

2.4. Statistical analysis

Participant characteristics are presented as either means and standard deviations or median and interquartile ranges.

In order to evaluate gait differences across age groups stratified by sex, we compared gait parameters between age categories (per decade) for men and women separately using a one-way analysis of variance test (ANOVA). In case of a significant effect of age category, we subsequently performed Tukey post hoc pairwise comparisons with Bonferroni correction of the p-values for multiple testing. All analyses were conducted in R version 4.3.1 with an alpha level of significance set at $p < 0.05$.

In order to graphically illustrate age-related changes, we provide sex-specific percentile curves generated with generalized additive models (GAMLSS, R package version 5.1–6). The age-trajectories were modeled using P-splines. The Bayesian information criterion selected the conditional distribution that balanced model complexity and goodness-of-fit. Model fits were inspected using diagnostic residual plots (worm plots and Q-Q plots).

Table 1

Measured parameters and their definitions as of the Physilog® user manual (9).

Gait Parameter	Definition and description
Speed [m/s]	Average forward walking speed, calculated in meters per second.
Cycle duration variability [%]	Coefficient of variation for cycle duration (discussed below) expressed as a percentage; calculated as $SD/mean * 100$.
Asymmetry [%]	Ratio of swing time comparing left and right feet; perfect symmetry yields 0%. Calculation as follows: $Asym. = \sqrt{\left(\frac{Swing_l - Swing_r}{0.5 * (Swing_l + Swing_r)} * 100\%\right)^2}$
Stride Length [m]	Distance between two successive footprints on the ground, measured from heel to heel of the same foot.
Cycle Duration [s]	Duration of one cycle, measured from heel strike to heel strike of the same foot.
Cadence [steps/min]	Number of cycles per minute.
Stance Phase [%]	Proportion of cycle duration during which the foot is in contact with the ground; typically around 60% of the gait cycle.
Double Support Phase [%]	Proportion of cycle duration during which both feet are simultaneously in contact with the ground.
Swing Phase [%]	Proportion of cycle duration during which the foot is in the air and not touching the ground; usually about 40% of the gait cycle.
Loading Phase [%]	Proportion of the stance phase between heel strike and the foot being flat on the ground.
Foot-Flat Phase [%]	Proportion of the stance phase when the foot is fully flat on the ground.
Pushing Phase [%]	Proportion of the stance phase between the foot being flat on the ground and the toe leaving the ground at take-off.

3. Results

A total of 629 participants performed the walking test. For the analysis, valid gait data from 545 participants (273 men and 272 women) were analyzed (see flow chart, Appendix 1). Participants were equally distributed across age groups from 20 to older than 80 years, with a minimum of 21 men and 30 women representing each decade. Participant baseline characteristics are presented in Table 2 by age group (i.e., per decade).

3.1. Gait parameters across the adult age range

The mean stride count per participant was 25.0 SD 2.9 (min 17, max 38). Gait parameters for each age group are presented in Table 3, separated by sex. Standardized mean differences are presented in Table 4. The results of the post-hoc pairwise comparisons of the ANOVA analysis for men and women (when applicable) are presented in Appendix 2.

The pattern of the different gait parameters over time (i.e. smoothed percentile curves) for men and women are visualized in Figs. 1 and 2, as well as supplementary Figures in the Appendix 3 and 4, providing a graphical representation of the changes across the lifespan.

3.1.1. Men

In men, we saw a significant effect of age decades for speed, stride length, cadence, cycle duration, the loading phase, and the foot-flat phase, with p-values ranging between 0.007 and < 0.001 (see Table 4). Looking at the post-hoc pairwise comparisons, we saw a significant lower gait speed in men aged 80 and above, compared to men between 30 and 39 ($p = 0.04$), between 40 and 49 ($p = 0.01$), between 50 and 59 ($p = 0.019$) and between 60 and 69 ($p = 0.006$). Cadence was significantly lower in men between 20 and 29 years old compared to men between 60 and 69 ($p = 0.004$) and between 70 and 79 ($p < 0.001$). Stride length was significantly lower in older men. We found significant differences between men aged 80 and above compared to men in all age categories under 70 years ($p < 0.001$ for all comparisons). Men aged 70 to 79 years had a significantly lower stride length than men in all younger age categories ($p < 0.02$). Cycle duration was significantly higher in men aged 20 to 29 and men between 60 and 69, and 70 and 79 ($p = 0.004$ and $p = 0.001$, respectively). The loading phase was significantly lower in men aged 80 and above compared to men between 40 and 49 and 60 and 69 ($p = 0.007$ and $p = 0.039$, respectively). Finally, the foot-flat phase was significantly higher in men aged 80 and above compared to men between 30 and 39 ($p = 0.009$),

between 40 and 49 ($p = 0.004$), and between 60 and 69 ($p = 0.018$). The results of pairwise comparisons are presented in Appendix 4.

3.1.2. Women

In women, we saw a significant effect of age decade on speed, stride length, cadence, cycle duration, the loading phase, the foot-flat phase, the cycle duration variability, and asymmetry with p-values ranging between 0.034 and < 0.001 (see Table 4). The post-hoc pairwise comparisons showed a significantly lower gait speed in women aged 80 and above compared to women below the age of 70 (p-values < 0.003). Furthermore, women between 70 and 79 years of age had a significantly lower gait speed than 50 to 59 year olds ($p = 0.043$). Stride length was significantly lower in women aged 80 and above compared to all younger age categories ($p < 0.007$), and in 70 to 79 year olds compared to women below 60 years of age ($p < 0.001$). Women between 60 and 69 had a lower stride length than women between 30 and 59 years of age ($p < 0.036$). The loading phase was significantly lower and the foot-flat phase was significantly higher in women aged 80 and above compared to all younger age categories ($p < 0.017$ and $p < 0.030$, respectively). The cycle duration variability was higher in women aged 80 and above compared to all age categories until 59 years ($p < 0.008$) and asymmetry was higher in the oldest age category compared to women until 49 years of age ($p < 0.019$). The results of pairwise comparisons are presented in Appendix 2.

4. Discussion

4.1. Main findings

In this study, we aimed to provide an overview of spatiotemporal gait parameters in healthy Swiss individuals aged 20 to 91 years, using an IMU-based gait analysis system. Our findings contribute to the understanding of gait patterns over the lifespan and may help identify risk factors for falls or indicators of disease.

We found that gait speed, stride length, cycle duration, cadence, loading, and foot-flat phase were significantly different across age categories for both men and women. In general, gait speed and stride length decreased with age, while cadence increased. The proportion of the stance phase and double support phase also evolved similarly across age groups.

Our results showed that gait patterns are relatively stable during early adulthood, with more pronounced changes occurring after the age of 60. Gait speed and stride length were found to be lower in older individuals. It is important to emphasize that the average gait speed for all

Table 2
Characteristics of participants (mean (SD)).

	Age group							
	Overall	20–29	30–39	40–49	50–59	60–69	70–79	80–91
N female	272	47	37	44	43	41	30	30
Age (years)	53.0 (19.2)	25.6 (3.1)	36.0 (3.2)	46.3 (2.7)	55.4 (2.9)	65.9 (2.7)	74.2 (2.6)	84.3 (2.4)
Height (cm)	165.5 (6.9)	166.7 (6.3)	169.6 (5.3)	168.6 (5.7)	167.4 (6.9)	161.7 (6.2)	161.2 (6.0)	160.3 (6.0)
Body mass (kg)	62.3 (8.5)	59.7 (8.6)	64.8 (8.5)	64.8 (7.4)	63.9 (8.5)	61.8 (8.5)	60.8 (7.9)	59.8 (9.1)
BMI (kg/m ²)	22.8 (2.7)	21.5 (2.7)	22.5 (2.9)	22.8 (2.4)	22.8 (2.2)	23.6 (2.8)	23.4 (3.1)	23.2 (2.9)
N male	273	44	45	39	38	42	44	21
Age (years)	52.9 (19.2)	25.7 (2.8)	35.2 (3.2)	46.1 (2.8)	55.9 (3.0)	64.8 (3.0)	75.1 (3.0)	84.5 (2.8)
Height (cm)	177.0 (7.5)	181.7 (6.8)	178.9 (6.7)	178.8 (5.9)	178.2 (6.2)	175.9 (7.3)	172.1 (7.1)	170.4 (6.7)
Body mass (kg)	76.7 (9.6)	77.3 (10.0)	77.9 (9.8)	76.7 (9.1)	79.3 (10.5)	76.5 (8.4)	73.8 (9.6)	75.2 (9.2)
BMI (kg/m ²)	24.5 (2.5)	23.4 (2.4)	24.3 (2.3)	24.0 (2.4)	24.9 (2.5)	24.7 (2.2)	24.9 (2.6)	25.9 (2.9)
N overall	545	91	82	83	81	83	74	51
Age (years)	52.9 (19.2)	25.6 (3.0)	35.5 (3.2)	46.2 (2.7)	55.6 (3.0)	65.4 (2.9)	74.7 (2.8)	84.4 (2.5)
Height (cm)	171.3 (9.3)	174.0 (10.0)	174.7 (7.6)	173.3 (7.7)	172.5 (8.5)	168.9 (9.8)	167.8 (8.6)	164.4 (8.0)
Body mass (kg)	69.6 (11.6)	68.3 (12.8)	71.9 (11.3)	70.3 (10.1)	71.2 (12.2)	69.2 (11.2)	68.7 (11.0)	66.1 (11.9)
BMI (kg/m ²)	23.6 (2.8)	22.4 (2.7)	23.5 (2.7)	23.3 (2.4)	23.8 (2.6)	24.2 (2.6)	24.3 (2.9)	24.3 (3.2)
Sex (%)								
Female	49.9	51.6	45.1	53.0	53.1	49.4	40.5	58.8
Male	50.1	48.4	54.9	47.0	46.9	50.6	59.5	41.2

Table 3
Gait parameters of female and male participants (mean (SD)).

	Age group							
	Overall	20–29	30–39	40–49	50–59	60–69	70–79	80–91
N female	272	47	37	44	43	41	30	30
Speed [m/s]	1.43 (0.15)	1.45 (0.14)	1.45 (0.13)	1.45 (0.12)	1.48 (0.14)	1.43 (0.14)	1.38 (0.14)	1.29 (0.20)
Variability [%]	2.44 (0.92)	2.30 (0.80)	2.18 (0.74)	2.35 (0.78)	2.13 (0.57)	2.51 (0.94)	2.82 (1.10)	3.09 (1.25)
Asymmetry [%]	2.15 (2.00)	1.98 (2.03)	1.42 (1.27)	1.82 (1.33)	2.18 (1.74)	2.63 (2.48)	1.82 (2.04)	3.48 (2.44)
Stride Length [m]	1.44 (0.13)	1.47 (0.10)	1.49 (0.11)	1.49 (0.10)	1.50 (0.11)	1.41 (0.10)	1.37 (0.09)	1.27 (0.14)
Cycle Duration [s]	1.02 (0.07)	1.03 (0.06)	1.04 (0.05)	1.04 (0.06)	1.02 (0.07)	1.00 (0.06)	1.01 (0.08)	1.00 (0.09)
Cadence [steps/min]	118.13 (7.76)	116.96 (6.57)	115.94 (6.25)	116.15 (6.55)	117.73 (8.31)	120.48 (7.02)	120.05 (8.45)	120.98 (10.38)
Stance [%]	60.64 (1.50)	60.73 (1.32)	60.91 (1.40)	60.81 (1.32)	60.47 (1.46)	60.75 (1.33)	60.71 (1.53)	59.96 (2.19)
Double Support [%]	21.25 (2.99)	21.42 (2.66)	21.82 (2.76)	21.59 (2.65)	20.91 (2.91)	21.50 (2.65)	21.28 (3.04)	19.87 (4.29)
Swing [%]	39.36 (1.50)	39.27 (1.32)	39.09 (1.40)	39.19 (1.32)	39.53 (1.46)	39.25 (1.33)	39.29 (1.53)	40.04 (2.19)
Loading [%]	13.73 (2.40)	14.23 (2.43)	14.59 (2.11)	13.56 (2.44)	14.68 (1.86)	13.71 (2.27)	13.20 (2.14)	11.30 (2.05)
Foot-flat [%]	49.92 (5.05)	48.68 (4.52)	49.34 (4.42)	50.02 (4.15)	48.95 (4.92)	49.56 (6.01)	50.14 (4.31)	54.09 (5.46)
Pushing [%]	36.36 (4.03)	37.09 (3.69)	36.07 (3.32)	36.42 (3.31)	36.37 (4.15)	36.79 (4.97)	36.66 (3.68)	34.61 (4.81)
N male	273	44	45	39	38	42	44	21
Speed [m/s]	1.43 (0.16)	1.43 (0.15)	1.43 (0.15)	1.46 (0.16)	1.45 (0.12)	1.46 (0.13)	1.40 (0.19)	1.31 (0.17)
Variability [%]	2.37 (0.87)	2.41 (0.77)	2.32 (0.71)	2.20 (0.65)	2.35 (0.90)	2.33 (0.88)	2.59 (1.25)	2.39 (0.69)
Asymmetry [%]	2.44 (2.22)	2.08 (1.89)	2.50 (2.02)	2.31 (2.51)	2.41 (2.67)	2.35 (1.87)	2.79 (2.29)	2.82 (2.46)
Stride Length [m]	1.51 (0.13)	1.55 (0.13)	1.53 (0.11)	1.55 (0.11)	1.54 (0.12)	1.52 (0.11)	1.44 (0.14)	1.37 (0.14)
Cycle Duration [s]	1.07 (0.06)	1.10 (0.06)	1.08 (0.06)	1.08 (0.06)	1.07 (0.05)	1.05 (0.06)	1.05 (0.07)	1.05 (0.06)
Cadence [steps/min]	112.63 (6.80)	109.20 (5.70)	111.94 (7.03)	111.83 (6.64)	111.99 (5.16)	114.48 (6.74)	115.47 (7.63)	114.28 (6.53)
Stance [%]	61.06 (1.38)	61.01 (1.43)	61.19 (1.48)	61.25 (1.54)	61.10 (1.09)	61.26 (1.30)	60.81 (1.56)	60.67 (0.96)
Double Support [%]	22.06 (2.78)	21.95 (2.89)	22.34 (2.97)	22.42 (3.08)	22.08 (2.23)	22.45 (2.60)	21.51 (3.10)	21.33 (2.04)
Swing [%]	38.94 (1.38)	38.99 (1.43)	38.81 (1.48)	38.75 (1.54)	38.90 (1.09)	38.74 (1.30)	39.19 (1.56)	39.33 (0.96)
Loading [%]	13.77 (2.55)	13.50 (2.58)	14.07 (2.08)	14.61 (2.94)	13.65 (2.46)	14.24 (2.55)	13.39 (2.78)	12.19 (1.42)
Foot-flat [%]	51.22 (4.47)	50.95 (4.31)	50.32 (4.47)	49.97 (4.27)	51.18 (3.36)	50.52 (4.12)	52.72 (5.08)	54.37 (4.56)
Pushing [%]	35.01 (3.47)	35.55 (2.90)	35.61 (3.83)	35.41 (3.21)	35.17 (3.02)	35.24 (3.23)	33.89 (3.47)	33.44 (4.70)

Abbreviations: N = number; m = metres; s = seconds; variability = cycle duration variability.

Table 4
Results of ANOVA test for the different gait parameters by age group.

	Female			–	Male		
	F	p	effect size (ges)		F	p	effect size (ges)
Speed [m/s]	6.390	< 0.001	0.126		3.017	0.007	0.064
Variability [%]	5.412	< 0.001	0.109		0.788	0.580	0.017
Asymmetry [%]	0.335	0.918	0.008		1.786	0.102	0.039
Stride Length [m]	21.979	< 0.001	0.332		9.714	< 0.001	0.180
Cycle Duration [s]	2.320	0.034	0.050		4.090	0.001	0.084
Cadence [steps/min]	2.892	0.010	0.061		4.402	< 0.001	0.090
Stance [%]	1.505	0.177	0.033		0.859	0.526	0.019
Double Support [%]	1.566	0.157	0.034		0.858	0.526	0.019
Swing [%]	1.505	0.177	0.033		0.859	0.526	0.019
Loading [%]	9.070	< 0.001	0.170		2.752	0.013	0.058
Foot-flat [%]	4.638	< 0.001	0.095		3.779	0.001	0.079
Pushing [%]	1.351	0.235	0.030		2.060	0.058	0.044

groups in our study was notably above the 1.0 m/s threshold, which has been suggested as an indicator of frailty [15].

4.2. Interpretation

Our observations suggest a compensatory increase in cadence among older age groups, possibly as an adaptive mechanism to maintain walking speed despite reductions in stride length. The observed changes in gait parameters with aging can be attributed to a range of interrelated factors. Age-related declines in muscle mass and strength, balance, and coordination [16–19], as well as alterations in the central and peripheral nervous systems [20,21], contribute to slower, more variable, and asymmetrical gait patterns. Additionally, a decline in motor unit number and function [22,23] exacerbates these effects. Joint health issues, such as osteoarthritis, limit movement and cause pain, leading to reduced gait speed and stride length [24]. Sensory declines, particularly in vision and proprioception, impact balance and spatial awareness, promoting more cautious gait patterns [25–27]. Cognitive factors, including multitasking abilities and attention, are also crucial, especially in older individuals, influencing gait dynamics [28]. Moreover, lifestyle choices and physical

activity levels significantly influence gait performance as one ages [29, 30]. These factors collectively affect stability and walking efficiency, often resulting in slower and less stable gait, characterized by increased cycle duration variability and asymmetry [31,32]. This variability is associated with a higher fall risk in older adults [1,33].

4.3. Comparison with other studies

Most studies have focused on either older or younger healthy adults. To the best of our knowledge, the Yishun Study by Lau et al. (2020) is the only other study reporting gait parameters over the entire lifespan [5]. Their study focused on a Southeast Asian population (n = 507, from Singapore), whereas our study provides an overview for Swiss individuals from mostly Caucasian origin. It has been described that gait varies between demographic groups [6,7]. Specifically, a study by Boulifard et al. (2019) found significant variations in gait across different demographics with African American and Hispanic participants having lower gait speeds than their white counterparts. Consistent with our findings, they reported that older participants had slower speeds than younger ones. Unlike our study, they observed generally

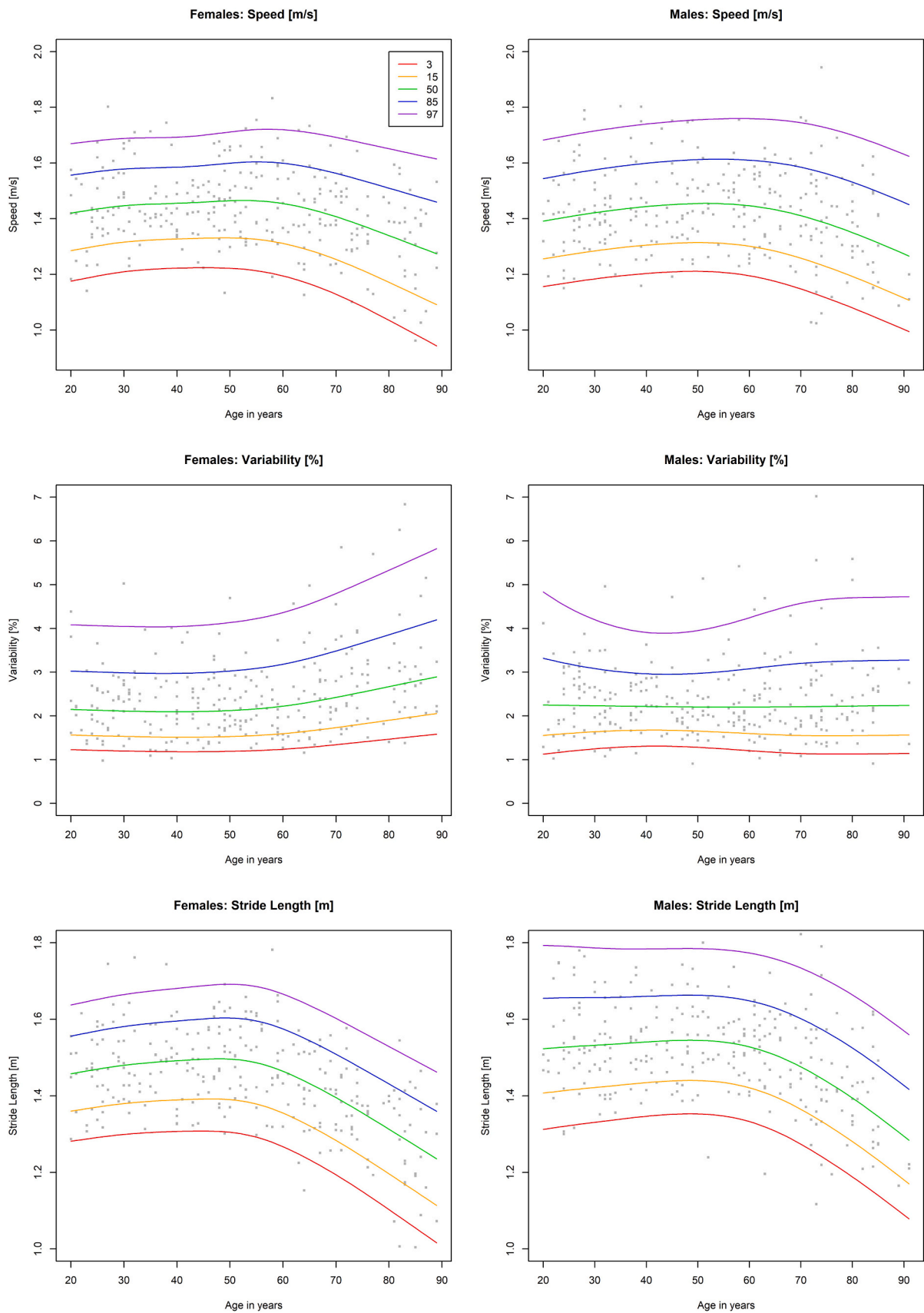


Fig. 1. Percentile curves for Speed, Variability, and Stride length.

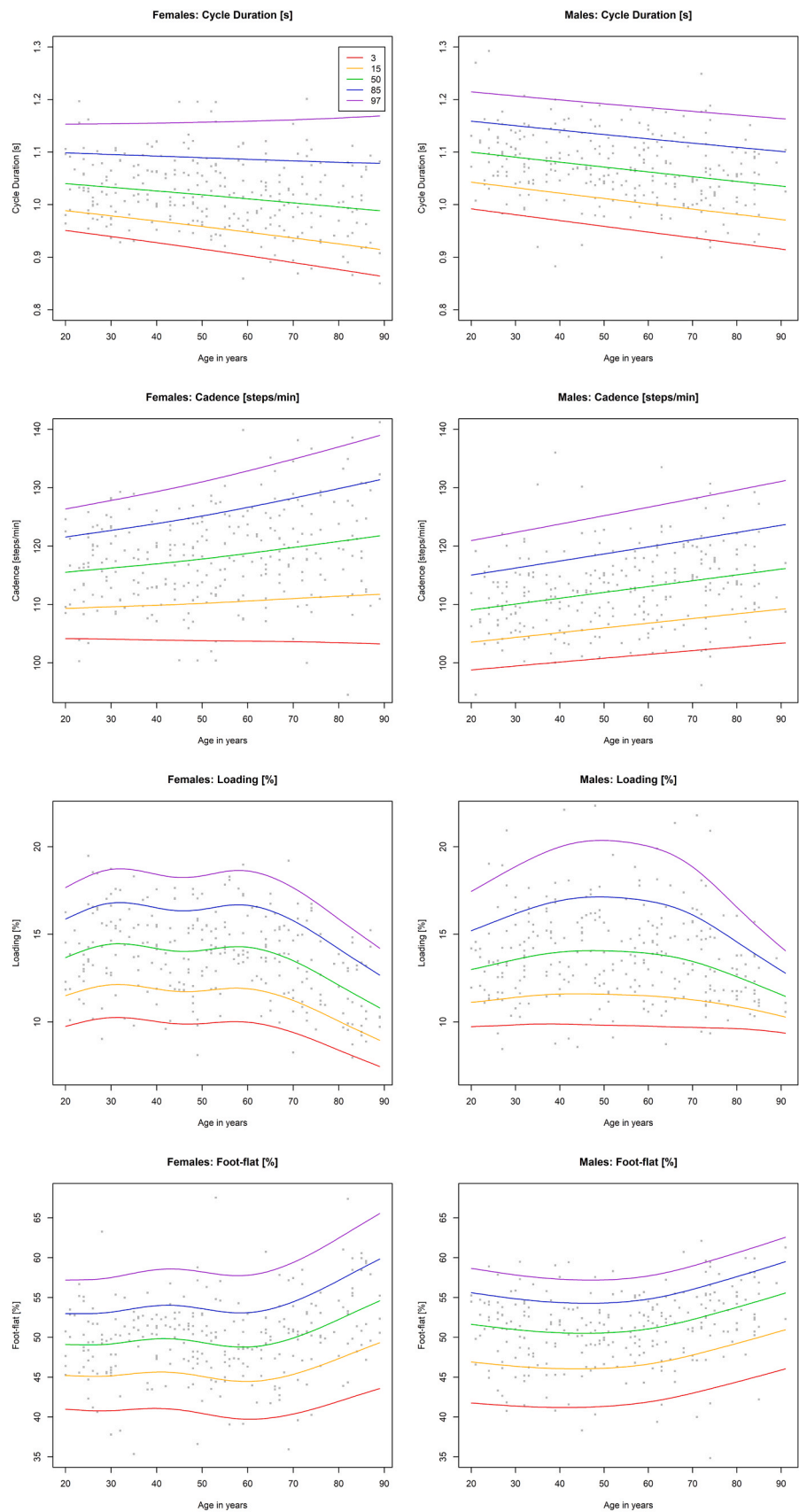


Fig. 2. Percentile curves for Cycle Duration, Cadence, Loading, and Foot-flat.

lower gait speeds in females compared to males [6]. A study from Hill et al. (2022) highlights that racial differences in gait speeds among females are influenced by both innate and modifiable factors, underscoring the importance of collecting norm values for different ethnic groups [7].

Both the study by Lau et al. (2020) and our study focused on “healthy individuals” (i.e. several diseases listed as exclusion criteria). We used a wearable sensor system to measure gait parameters, which allowed for unobtrusive data collection in a natural walking environment. The Yishun Study employed the GAITRite electronic walkway system, a widely used tool in gait analysis (often considered as “the golden standard”). This system requires participants to walk on a special mat with integrated sensors to record gait parameters. Although both methods are valid for gait analysis, our approach allowed for a more naturalistic assessment and longer walking distances (i.e. two times 20 m per attempt) [5].

Interestingly, for both females and males, participants in our study consistently showed higher gait speeds, longer step/stride lengths, and increased cadence across all age groups, compared to the Asian study. Both studies involved healthy participants who were instructed to walk at their habitual pace. A notable distinction is that participants in our study wore shoes, while those in the Yishun Study walked barefoot [5]. We are not aware of any studies explicitly linking these observed differences to whether participants are barefoot or shod. Therefore, the variations in gait characteristics observed in our study may be attributable to ethnic differences and/or disparities in test conditions.

In our study, we found significant differences in gait speed, stride length, cycle duration, cadence, loading, and foot-flat phase across age categories for both men and women. Gait speed and stride length generally decreased with age, while cadence increased. These changes were more pronounced in older individuals, particularly after the age of 60. The Yishun Study also reported age-related declines in gait speed and stride length, as well as increases in stance time and double support time, which are consistent with our findings [5]. Unlike our study, the Yishun Study did not report an increase in cadence with age. This difference might be attributed to the different populations studied, an Asian urban population versus a Swiss suburban and rural population. The differences observed between the two studies highlight the importance of considering population-specific factors when interpreting gait parameters and their potential clinical implications.

There are further studies reporting a decline in gait speed and stride length with aging [31,34]. The observed changes in cycle duration variability and asymmetry in our study are in line with previous research [4], which reported an increase in cycle duration variability with age and the association between age and gait asymmetry [35]. The results of our study contribute to the understanding of gait changes throughout the lifespan and can be compared to other populations to assess the generalizability of our findings.

4.4. Strengths and limitations

Strengths of this study include the large sample size and the wide age range of participants, which allowed for the investigation of gait changes at all ages between 20 and 91 years. Furthermore, the use of a reliable and validated system for gait assessment (Physilog 5) ensured accurate and consistent data collection [13,14].

The primary limitation of this study is that only one trial was captured per participant. However, the analysis was based on an average of 25 strides per participant, which is sufficient according to the literature [36].

Our study is founded on robust data derived from Swiss individuals, predominantly of Caucasian origin. While inclusive and diverse research is preferred, such inclusivity was not feasible in our context due to challenges in recruiting a balanced number of participants from various ethnic groups. Nonetheless, focusing primarily on one ethnic group, as our study does, can yield detailed and specific insights pertinent to the

group in question.

5. Conclusion

Our study provides valuable insights into the changes in gait parameters from 20 to over 90 years of age, revealing a decline in gait speed and stride length, and an increase in cadence, cycle duration variability, and asymmetry with aging. These findings provide valuable reference values for clinical practice and can inform interventions aimed at maintaining or improving gait performance in older adults. Targeted exercise programs focusing on strength, balance, and coordination could be particularly effective in preserving normal gait characteristics and reducing fall risk.

CRedit authorship contribution statement

Roland Rössler: Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Jonathan Wagner:** Data curation, Investigation, Project administration, Validation, Writing – review & editing. **Raphael Knaier:** Data curation, Investigation, Project administration, Validation. **Nikki Rommers:** Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Reto W. Kressig:** Validation, Writing – review & editing. **Arno Schmidt-Trucksäss:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing. **Timo Hinrichs:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

None of the authors declares a conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.gaitpost.2024.01.005.

References

- [1] B.E. Maki, Gait changes in older adults: predictors of falls or indicators of fear, *J. Am. Geriatr. Soc.* 45 (3) (1997) 313–320. (<https://www.ncbi.nlm.nih.gov/pubmed/9063277>).
- [2] K. Thaler-Kall, A. Peters, B. Thorand, E. Grill, C.S. Autenrieth, A. Horsch, et al., Description of spatio-temporal gait parameters in elderly people and their association with history of falls: results of the population-based cross-sectional KORA-Age study, *BMC Geriatr.* 15 (2015) 32. (<https://www.ncbi.nlm.nih.gov/pubmed/25880255>).
- [3] F. Ayoubi, C.P. Launay, A. Kabeshova, B. Fantino, C. Annweiler, O. Beauchet, The influence of fear of falling on gait variability: results from a large elderly population-based cross-sectional study, *J. Neuroeng. Rehabil.* 11 (2014) 128. (<https://www.ncbi.nlm.nih.gov/pubmed/25168467>).
- [4] N. Herssens, E. Verbecque, A. Hallemans, L. Vereeck, V. Van Rompaey, W. Saeys, Do spatiotemporal parameters and gait variability differ across the lifespan of healthy adults? A systematic review, *Gait Posture* 64 (2018) 181–190. (<https://www.ncbi.nlm.nih.gov/pubmed/29929161>).
- [5] L.K. Lau, S.L. Wee, W.J.B. Pang, K.K. Chen, K. Abdul Jabbar, P.L.K. Yap, et al., Reference values of gait speed and gait spatiotemporal parameters for a south east asian population: the Yishun Study, *Clin. Inter. Aging* 15 (2020) 1753–1765. (<https://www.ncbi.nlm.nih.gov/pubmed/33061327>).
- [6] D.A. Boulifard, E. Ayers, J. Verghese, Home-based gait speed assessment: normative data and racial/ethnic correlates among older adults, *J. Am. Med. Dir. Assoc.* 20 (10) (2019) 1224–1229. (<https://www.ncbi.nlm.nih.gov/pubmed/31395494>).

- [7] C.N. Hill, W. Reed, D. Schmitt, S.M. Arent, L.P. Sands, R.M. Queen, Factors contributing to racial differences in gait mechanics differ by sex, *Gait Posture* 95 (2022) 277–283. (<https://www.ncbi.nlm.nih.gov/pubmed/33658154>).
- [8] K. Carroll, R. Kennedy, V. Koutoulas, M. Bui, C. Kraan, Validation of shoe-worn Gait Up Physilog® 5 wearable inertial sensors in adolescents, *Gait Posture* 91 (2022) 19–25.
- [9] C.J. Hass, P. Malczak, J. Nocera, E.L. Stegemoller, A. Wagle Shukla, I. Malaty, et al., Quantitative normative gait data in a large cohort of ambulatory persons with Parkinson's disease, *PLoS One* 7 (8) (2012) e42337. (<https://www.ncbi.nlm.nih.gov/pubmed/22879945>).
- [10] P.A. Goldie, T.A. Matyas, O.M. Evans, Gait after stroke: initial deficit and changes in temporal patterns for each gait phase, *Arch. Phys. Med. Rehab.* 82 (8) (2001) 1057–1065. (<https://www.ncbi.nlm.nih.gov/pubmed/11494185>).
- [11] L. Comber, R. Galvin, S. Coote, Gait deficits in people with multiple sclerosis: a systematic review and meta-analysis, *Gait Posture* 51 (2017) 25–35. (<https://www.ncbi.nlm.nih.gov/pubmed/27693958>).
- [12] J. Wagner, R. Knaier, D. Infanger, K. Arbeev, M. Briel, T. Dieterle, et al., Functional aging in health and heart failure: the COMpLETE Study, *BMC Cardiovasc. Disord.* 19 (1) (2019) 180. (<https://www.ncbi.nlm.nih.gov/pubmed/31362698>).
- [13] F. Dadashi, B. Mariani, S. Rochat, C.J. Bula, B. Santos-Eggimann, K. Aminian, Gait and foot clearance parameters obtained using shoe-worn inertial sensors in a large-population sample of older adults, *Sensors (Basel)* 14 (1) (2013) 443–457. (<https://www.ncbi.nlm.nih.gov/pubmed/24379049>).
- [14] B. Mariani, C. Hoskovec, S. Rochat, C. Bula, J. Penders, K. Aminian, 3D gait assessment in young and elderly subjects using foot-worn inertial sensors, *J. Biomech.* 43 (15) (2010) 2999–3006. (<https://www.ncbi.nlm.nih.gov/pubmed/20656291>).
- [15] L.P. Fried, C.M. Tangen, J. Walston, A.B. Newman, C. Hirsch, J. Gottdiener, et al., Frailty in older adults: evidence for a phenotype, *J. Gerontol. A Biol. Sci. Med. Sci.* 56 (3) (2001) M146–M156. (<https://www.ncbi.nlm.nih.gov/pubmed/11253156>).
- [16] R.W. Bohannon, Reference values for extremity muscle strength obtained by hand-held dynamometry from adults aged 20 to 79 years, *Arch. Phys. Med. Rehab.* 78 (1) (1997) 26–32. (<https://www.ncbi.nlm.nih.gov/pubmed/9014953>).
- [17] U. Granacher, A. Gollhofer, T. Hortobagyi, R.W. Kressig, T. Muehlbauer, The importance of trunk muscle strength for balance, functional performance, and fall prevention in seniors: a systematic review, *Sports Med.* 43 (7) (2013) 627–641. (<https://www.ncbi.nlm.nih.gov/pubmed/23568373>).
- [18] D.P. LaRoche, K.A. Cremin, B. Greenleaf, R.V. Croce, Rapid torque development in older female fallers and nonfallers: a comparison across lower-extremity muscles, *J. Electromyogr. Kinesiol.* 20 (3) (2010) 482–488. (<https://www.ncbi.nlm.nih.gov/pubmed/19782579>).
- [19] T.W. Auyeung, S.W.J. Lee, J. Leung, T. Kwok, J. Woo, Age-associated decline of muscle mass, grip strength and gait speed: a 4-year longitudinal study of 3018 community-dwelling older Chinese, *Geriatr. Gerontol. Int.* 14 (2014) 76–84.
- [20] M.L. Callisaya, R. Beare, T.G. Phan, L. Blizzard, A.G. Thrift, J. Chen, et al., Brain structural change and gait decline: a longitudinal population-based study, *J. Am. Geriatr. Soc.* 61 (7) (2013) 1074–1079. (<https://www.ncbi.nlm.nih.gov/pubmed/23796055>).
- [21] K. Jahn, A. Zwergal, R. Schniepp, Gait disturbances in old age: classification, diagnosis, and treatment from a neurological perspective, *Dtsch Arztebl Int.* 107 (17) (2010) 306–15; quiz 316. (<https://www.ncbi.nlm.nih.gov/pubmed/20490346>).
- [22] D.J. Clark, R.M. Pojednic, K.F. Reid, C. Patten, E.P. Pasha, E.M. Phillips, Longitudinal decline of neuromuscular activation and power in healthy older adults, *J. Gerontol. Ser. A: Biol. Sci. Med. Sci.* 67(12) (2012) 1419–1425. (<https://doi.org/10.1093/gerona/gls197>).
- [23] H. Degens, T.M. Maden-Wilkinson, A. Ireland, M.T. Korhonen, H. Suominen, A. Heinonen, et al., Relationship between ventilatory function and age in master athletes and a sedentary reference population, *Age (Dordr.)* 35 (3) (2013) 1007–1015. (<https://www.ncbi.nlm.nih.gov/pubmed/22544616>).
- [24] S.P. Messier, Osteoarthritis of the knee and associated factors of age and obesity: effects on gait, *Med Sci. Sports Exerc.* 26 (12) (1994) 1446–1452. (<https://www.ncbi.nlm.nih.gov/pubmed/7869877>).
- [25] A. Hallemans, E. Ortibus, F. Meire, P. Aerts, Low vision affects dynamic stability of gait, *Gait Posture* 32 (4) (2010) 547–551. (<https://www.ncbi.nlm.nih.gov/pubmed/20801658>).
- [26] J.R. Wingert, C. Welder, P. Foo, Age-related hip proprioception declines: effects on postural sway and dynamic balance, *Arch. Phys. Med. Rehab.* 95 (2) (2014) 253–261. (<https://www.ncbi.nlm.nih.gov/pubmed/23994251>).
- [27] M. Henry, S. Baudry, Age-related changes in leg proprioception: implications for postural control, *J. Neurophysiol.* 122 (2) (2019) 525–538. (<https://www.ncbi.nlm.nih.gov/pubmed/31166819>).
- [28] N. Theill, M. Martin, V. Schumacher, S.A. Bridenbaugh, R.W. Kressig, Simultaneously measuring gait and cognitive performance in cognitively healthy and cognitively impaired older adults: the Basel motor-cognition dual-task paradigm, *J. Am. Geriatr. Soc.* 59 (6) (2011) 1012–1018. (<https://www.ncbi.nlm.nih.gov/pubmed/21649627>).
- [29] K.A. Boyer, T.P. Andriacchi, G.S. Beaupre, The role of physical activity in changes in walking mechanics with age, *Gait Posture* 36 (1) (2012) 149–153. (<https://www.ncbi.nlm.nih.gov/pubmed/22445586>).
- [30] D. Ciprandi, F. Bertozzi, M. Zago, C.L.P. Ferreira, G. Boari, C. Sforza, et al., Study of the association between gait variability and physical activity, *Eur. Rev. Aging Phys. Act.* 14 (1) (2017) 19. (<https://www.ncbi.nlm.nih.gov/pubmed/29177018>).
- [31] J.H. Hollman, E.M. McDade, R.C. Petersen, Normative spatiotemporal gait parameters in older adults, *Gait Posture* 34 (1) (2011) 111–118. (<https://www.ncbi.nlm.nih.gov/pubmed/21531139>).
- [32] J. Verghese, C. Wang, R.B. Lipton, R. Holtzer, X. Xue, Quantitative gait dysfunction and risk of cognitive decline and dementia, *J. Neurol. Neurosurg. Psychiatry* 78 (9) (2007) 929–935. (<https://www.ncbi.nlm.nih.gov/pubmed/17237140>).
- [33] S.R. Lord, H.B. Menz, A. Tiedemann, A physiological profile approach to falls risk assessment and prevention, *Phys. Ther.* 83 (3) (2003) 237–252. (<https://www.ncbi.nlm.nih.gov/pubmed/12620088>).
- [34] S. Studenski, S. Perera, K. Patel, C. Rosano, K. Faulkner, M. Inzitari, et al., Gait speed and survival in older adults, *JAMA* 305 (1) (2011) 50–58. (<https://www.ncbi.nlm.nih.gov/pubmed/21205966>).
- [35] O. Beauchet, G. Allali, C. Annweiler, S. Bridenbaugh, F. Assal, R.W. Kressig, et al., Gait variability among healthy adults: low and high stride-to-stride variability are both a reflection of gait stability, *Gerontology* 57 (5) (2011) 435–440. (<https://doi.org/10.1159/000322198>).
- [36] D. Kroneberg, M. Elshehabi, A.C. Meyer, K. Otte, S. Doss, F. Paul, et al., Less is more - estimation of the number of strides required to assess gait variability in spatially confined settings, *Front. Aging Neurosci.* 10 (2018) 435. (<https://www.ncbi.nlm.nih.gov/pubmed/30719002>).